

DSA Day 14: Deque & Priority Queue

Book: Data Structures Through C in Depth - Ch4 §4.4-4.5 · pp 123-139 | Code in Python

Overview

A deque allows insert/delete at both ends. A priority queue serves the highest-priority element first (often a heap).

Key concepts to master

- Insert/delete both ends
- Priority ordering
- `collections.deque`
- `heapq` for priority

Python implementation

```
from collections import deque
import heapq
d=deque(); d.appendleft(1); d.append(2); d.pop(); d.popleft()
pq=[]; heapq.heappush(pq, (2, 'b')); heapq.heappush(pq, (1, 'a')); heapq.heappop(pq)
```

Complexity

deque ends $O(1)$; heap push/pop $O(\log n)$.

Common pitfalls

- Using `list.pop(0)` ($O(n)$) instead of deque
- Forgetting `heapq` is a min-heap

Practice

- Run both-end operations
- Dequeue by priority

